

Nanoscale amorphization of poly(triarylamine) for efficient and stable inverted perovskite photovoltaics

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Perovskite solar modules require hole-selective layers that combine efficient charge extraction, interfacial uniformity and scalable processing. Poly(triarylamine) (PTAA) is widely used in high-performance inverted perovskite photovoltaics, but its nanoscale crystallization and aggregation on indium tin oxide can disrupt film continuity, increase interfacial recombination and limit module stability. Here we show that 4-fluorobenzylphosphonic acid (4FBPA) modifies the surface of indium tin oxide to induce nanoscale amorphization of PTAA, forming a uniform sub-10-nm hole-selective layer. The molecule binds to indium tin oxide through a dehydration reaction, tunes the work function and surface free energy of the photoanode, and improves energy-level alignment with PTAA. The resulting amorphous PTAA film shows enhanced conductivity and hole extraction, suppresses non-radiative recombination at the buried interface and promotes more uniform perovskite growth. Inverted perovskite solar cells reach a power conversion efficiency of 26.63%, while blade-coated modules achieve a certified quasi-steady-state efficiency of 23.01%. The modules retain 95.9% of their initial efficiency after 2,600 hours of maximum-power-point operation under 1-sun illumination at $65 \pm 5^\circ\text{C}$ in nitrogen. These results identify nanoscale amorphization of polymeric hole-selective layers as a route to efficient and stable inverted perovskite photovoltaics.

Inverted perovskite solar cells (p-i-n PSCs) with certified efficiencies exceeding 26% have surpassed their conventional (n-i-p) counterparts^{1–3}, emerging as the mainstream structure and front-runners in photovoltaic technology. These p-i-n PSCs are notable for their superior stability^{4–6}, large-area scalability^{7–9} and integrability with tandem architectures^{10–12}. Despite the rapid efficiency breakthrough of p-i-n PSCs, challenges remain in the efficiency and long-term stability of perovskite solar modules (PSMs). Issues caused by inhomogeneity and

defects within perovskite films during the scaling-up process have been effectively addressed through various strategies^{13–15}. However, unlike the automated techniques of thermal evaporation or atomic layer deposition for electron-selective layer deposition^{16–18}, hole-selective layers (HSLs) still rely on spin-coating or blade-coating methods. These processes are less controllable and face challenges in achieving high-quality and uniform films^{19–21}, leading to insufficient charge extraction and detrimental non-radiative recombination at the perovskite/HSL interface²².

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Poly(triarylamine), a polymeric hole transport material known as PTAA, exhibits inherent stability advantages^{23–25}, yet the efficiency of hole-selective polymer-based PSCs remains inferior to that of devices using organic small molecules or inorganic materials^{21,26,27}. Furthermore, due to the low conductivity of PTAA and the need to prevent additive induced degradation of the perovskite layer, undoped PTAA is generally used as an HSL in p–i–n architecture¹, requiring the HSL to be thin (~10 nm)²⁵ to minimize series resistance and fully cover the indium tin oxide (ITO) photoanode²⁸. We found that when PTAA was directly deposited onto the ITO surface, it exhibited partial crystallization along with a certain degree of aggregation. These behaviours influence the morphology and uniformity of the PTAA film, generating defects at the PTAA/perovskite interfaces and increasing non-radiative recombination, ultimately posing challenges for the large-scale fabrication of PSCs²⁹. Therefore, achieving uniform coverage of amorphous PTAA without aggregation is crucial for improving the performance of p–i–n PSCs and PSMs, which have yet to be thoroughly investigated.

In this work, we introduced an amphiphilic molecule, 4-fluorobenzylphosphonic acid (4FBPA), onto the ITO surface to modulate its surface properties and regulate the morphological behaviour of PTAA. Our characterization results demonstrate that the incorporation of 4FBPA alters the surface free energy of ITO, exhibiting the potential to modulate the PTAA film formation process. Moreover, the grazing incidence wide-angle X-ray scattering (GIWAXS) results show that the 4FBPA-modified substrate suppresses the crystallization of PTAA and promotes the formation of a uniform, amorphous PTAA film. Therefore, we used the 4FBPA to tailor the surface properties of ITO, enabling the amorphous growth of PTAA to enhance uniformity in inverted PSCs and PSMs (Fig. 1a). The resulting amorphous and uniform PTAA exhibits enhanced conductivity and hole extraction, addressing the limitations associated with PTAA-based PSCs and PSMs. Through this tailored molecular blade-coating process, small-area PSCs achieved an impressive efficiency of 26.63%, while PSMs obtained a quasi-steady-state (QSS) certified efficiency of 23.01% (the laboratory-tested efficiency is 23.64%) with excellent stability. The PSMs retained over 95.9% of their initial power conversion efficiencies (PCEs) (versus 56.1% for the control) over 2,600 hours under maximum-power-point tracking conditions (1-sun illumination, 65 ± 5 °C and nitrogen atmosphere). Moreover, the PSMs maintained over 90% of their initial PCEs (versus 41.9% for the control) after 1,500 hours under damp-heat conditions (85 °C and 85% humidity).

Phosphonic acid-engineered ITO photoanode

Here we use two organic phosphonic acids (OPAs), benzylphosphonic acid (BPA) and 4FBPA, to tailor the surface properties of the ITO photoanode (Fig. 1b). The molecular structures are depicted in Supplementary Fig. 1. Density functional theory (DFT) calculations reveal that the two hydroxyl groups within OPA molecules tend to bind to the ITO surface by forming ester groups through a dehydration reaction (Fig. 1c and Supplementary Fig. 2). Notably, the DFT results demonstrate that the work function variations relative to bare ITO are 0.13 eV (BPA–ITO) and 0.41 eV (4FBPA–ITO), indicating the potential of BPA and 4FBPA as effective surface regulators for fine-tuning ITO properties. Moreover, the binding energy between PTAA and 4FBPA–ITO is enhanced compared to other samples, which is beneficial for improving the interfacial stability of PSCs or PSMs.

We performed X-ray photoelectron spectroscopy to probe the interaction between ITO and OPA molecules (Supplementary Fig. 3). In the survey spectra (Supplementary Fig. 3b,c), the detectable signals of F 1s and P 2s confirm the presence of OPA molecules on the ITO surface. A slight peak shift for the Sn 4s and In 4s (Supplementary Fig. 4), as well as a decrease in the Sn–O or In–O peak intensities of BPA–ITO and 4FBPA–ITO, can be attributed to the chemical bond formation between ITO and the OPAs, aligning with the previous DFT analysis. Specifically, the O 1s signal provides additional evidence for this interfacial reaction.

Compared with bare ITO (Supplementary Fig. 3d), the OPA-modified samples (Supplementary Fig. 3e,f) exhibit a decrease in the intensity of Sn–O/In–O from 64% to 57% and an increase in the intensity of O=P from 27% to 37%, further confirming interfacial chemical bonding between OPA molecules and ITO photoanode. A schematic of the OPA–ITO reaction process is shown in Fig. 1b. Additional details are discussed in Supplementary Note 1.

To evaluate the energetics properties of the OPA–ITOs, we performed UV photoelectron spectroscopy (Supplementary Fig. 5) to acquire the surface electronic structure and calculate the corresponding work function values of the bare ITO and OPA–ITOs. The work function of 4FBPA–ITO is well-aligned with the valence band of PTAA (Supplementary Fig. 5e), which is beneficial for reducing the energy barrier and facilitating hole extraction at the ITO/HSL interface.

Amorphous PTAA and its impact on perovskites

We then conducted GIWAXS to identify the crystallinity of PTAA molecules on the resulting ITO substrates (Fig. 2a–c), with the corresponding 1D scattering profile shown in Supplementary Fig. 6. In the bare ITO/PTAA and BPA–ITO/PTAA samples, additional peaks with a preferred orientation are observed between 10° and 20° (corresponding to $0.90 \text{ \AA}^{-1}$ and $1.06 \text{ \AA}^{-1}$). In contrast, PTAA on 4FBPA–ITO (Fig. 2c) exhibits only the typical ITO peaks (Supplementary Fig. 6a) at around 30° , indicating the amorphous state growth of PTAA.

We further used Kelvin probe force microscopy (KPFM) to investigate the impact of OPAs on the surface potential of ITO as shown in Supplementary Fig. 7a–c, with the corresponding contact potential difference (CPD) distributions extracted in Supplementary Fig. 7d. These results indicate that BPA–ITO and 4FBPA–ITO exhibit different surface potential values from bare ITO, with uniform spatial distribution. In addition, we employed potassium ferricyanide as an external redox probe to evaluate the interfacial blocking effect of OPAs on the ITO substrate. The cyclic voltammetry responses of bare ITO, BPA–ITO and 4FBPA–ITO were recorded under identical conditions (Supplementary Fig. 8). The results show that the probe currents of BPA–ITO and 4FBPA–ITO are nearly identical. Combined with the previous KPFM results, it suggests that BPA and 4FBPA exhibit similar surface coverage on ITO and both achieve nearly complete coverage of the ITO surface.

The corresponding topography of PTAA deposited on bare ITO and OPA–ITO substrates is illustrated in Fig. 2d–f. The results demonstrate that there are clear variations in the morphology of PTAA on different substrates. An obvious dark region and particles were observed when PTAA was deposited on both ITO and BPA–ITO substrates (Supplementary Figs. 9 and 10), suggesting the presence of surface voids and partial aggregation. By contrast, there was no aggregation of PTAA on the 4FBPA–ITO substrate. We also studied the effect of OPAs on the electrical behaviours of PTAA using conductive atomic force microscopy. The amorphous and non-aggregated PTAA on 4FBPA–ITO shows higher current values than the partial crystalline PTAA on both bare ITO and BPA–ITO, indicating superior conductivity in the 4FBPA–ITO/amorphous PTAA structure (Fig. 2g–i and Supplementary Fig. 11).

The contact-angle measurements (Supplementary Fig. 12) reveal that the altered surface free energy of ITO is one of the factors affecting PTAA molecule crystallinity (Supplementary Note 2 and Supplementary Table 1). The lower surface free energy of 4FBPA-modified ITO favours the formation of amorphous PTAA films. Specifically, the reduced surface free energy of the substrate weakens the interfacial interaction between PTAA molecules and the substrate, increases the heterogeneous nucleation barrier of PTAA on the surface and enables PTAA chains to retain a disordered and entangled conformation. The above approaches demonstrate that 4FBPA molecules can fine-tune the energetics of the ITO photoanode, reduce series resistance and enable high-performance inverted PSCs and PSMs.

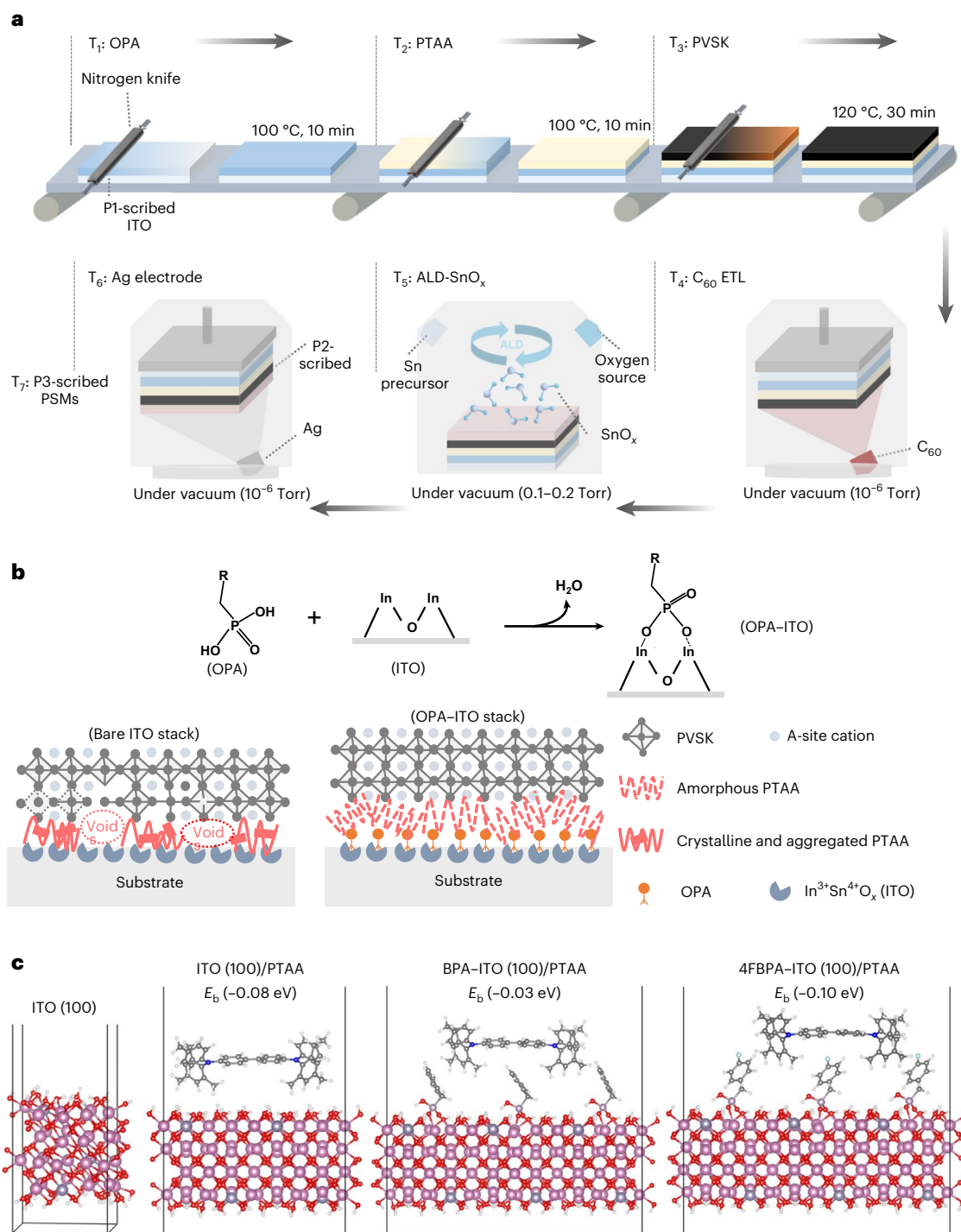


Fig. 1 | Impact of OPA molecules on the ITO photoanode and HSL (PTAA).

a, Schematic illustration of the typical processing steps (T₁ to T₇) for p–i–n PSMs. **b**, Schematic of the OPA–ITO dehydration reaction and the resulting PTAA growth: crystalline and aggregated on bare ITO, and amorphous on OPA–ITO. **c**, Optimized slab structures of OPA–ITO (100) coordinated by PTAA molecules,

in comparison to pristine ITO (100), along with the corresponding calculated binding energy (*E_b*) values. These calculations were performed at the generalized gradient approximation level of theory using the Perdew–Burke–Ernzerhof functional and van der Waals corrections. ETL, electron transport layer; PVSK, perovskite.

We first investigated the influence of different 4FBPA concentrations on PSC performance and identified an optimal concentration of 0.4 mg ml⁻¹ (Supplementary Fig. 13). Meanwhile, we also evaluated the feasibility of using 4FBPA directly as the HSL (Supplementary Fig. 14) and Li-TFSI-doped PTAA (Supplementary Fig. 15). These results indicate the rationale for selecting the 4FBPA/PTAA (undoped) HSL over a self-assembled monolayer HSL, as it simultaneously enhances the efficiency of PTAA-based devices while maintaining their superior

stability. Further details are provided in Supplementary Note 3. Unless otherwise stated, we primarily evaluate the impact of two configurations on inverted PSCs: bare ITO with crystalline PTAA as the control and 4FBPA–ITO with amorphous PTAA as the target. The perovskite composition is Cs_{0.05}FA_{0.95}PbI₃ with 10% excess PbI₂. Supplementary Fig. 16 compares the morphologies of the perovskite films on the control and target substrates using scanning electron microscopy (SEM), and the target film exhibits better uniformity and larger grain size. The narrow

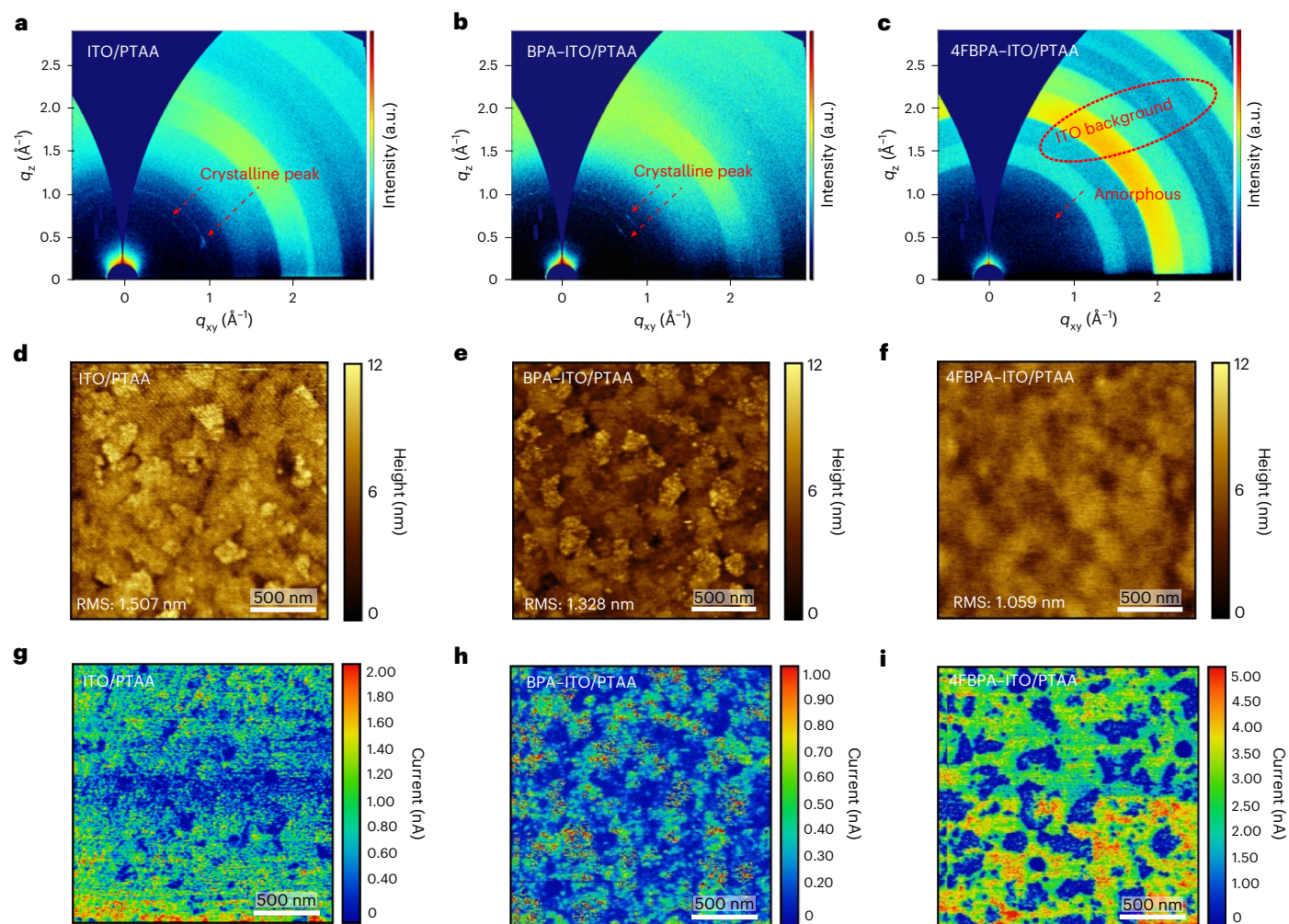


Fig. 2 | Physical characteristics of PTAA films. **a–c**, Two-dimensional GIWAXS patterns of ITO/PTAA (**a**), BPA-ITO/PTAA (**b**) and 4FBPA-ITO/PTAA (**c**). Crystalline peaks of PTAA are indicated by red arrows, while the bright rings originate from the bare ITO background (red dashed circle region). **d–f**, AFM

images of the topography of PTAA deposited on ITO (**d**), BPA-ITO (**e**) and 4FBPA-ITO (**f**). **g–i**, Conductive atomic force microscopy current mapping images for ITO/PTAA (**g**), BPA-ITO/PTAA (**h**) and 4FBPA-ITO/PTAA (**i**). RMS, root-mean-square roughness.

full-width at half-maximum for the target perovskite in X-ray diffraction patterns (Supplementary Fig. 17) and steady-state photoluminescence (PL) (Supplementary Fig. 18), as well as its correlation with SEM results, indicates better crystal quality with reduced defects³⁰.

We further measured time-resolved PL and femtosecond transient absorption spectroscopy of the perovskite films on different substrates. The fitted time-resolved PL decay of the target sample (Supplementary Fig. 19 and Supplementary Table 2) shows a faster decay time component τ_1 , which is related to an efficient charge carrier transfer from the perovskite to the 4FBPA-ITO³¹. From femtosecond transient absorption spectroscopy measurements the target sample exhibits a longer-lived ground-state bleach signal (Supplementary Fig. 20 and Supplementary Table 3) compared to the control sample, indicating increased photogenerated carriers. In this case, the slower ground-state bleach recovery in the target sample reveals an effectively suppressed non-radiative recombination process.

Device performance of perovskite photovoltaics

According to the process shown in Fig. 1a, we conducted a comprehensive evaluation of device characteristics using the tailored ITO within a typical inverted architecture, that is, ITO/PTAA/Cs_{0.05}FA_{0.95}PbI₃/fullerene (C₆₀)/SnO_x/Ag. Figure 3a presents the cross-sectional SEM image of the device structure. Cryogenic scanning transmission electron

microscopy (cryo-STEM) under low-dose conditions was used to characterize the 4FBPA-ITO/PTAA/perovskite stack (Fig. 3b). As shown in Fig. 3c, the PTAA layer has a thickness of less than 10 nm and exhibits an amorphous state, consistent with the GIWAXS results. PTAA films grown on bare ITO and BPA-ITO substrates show similar thickness in the range of 5–10 nm (Supplementary Fig. 21a,b). Additionally, electron energy-loss spectroscopy (EELS) provides evidence for the presence of 4FBPA molecules at the ITO interface through the detection of fluorine (F) (Fig. 3d and Supplementary Fig. 21c). Figure 3e compares the photovoltaic parameters of the resulting PSCs under standard 1-sun illumination, with detailed photovoltaic parameters provided in Supplementary Table 4. The efficiency and stability of BPA-ITO devices are presented in Supplementary Fig. 22. Notably, BPA-ITO devices show a trend similar to that of the control devices. The champion target device exhibited a laboratory-measured efficiency of 26.63% and a reverse scan efficiency of 25.81% measured at an accredited photovoltaics laboratory (NREL PV Performance Group) (Supplementary Fig. 23). Supplementary Fig. 24 presents the stabilized power outputs of the control and target devices, which are 24.81% and 26.45%, respectively. The statistical distribution of photovoltaic parameters for 40 target devices from the same batch exhibited good reproducibility (Fig. 3f and Supplementary Fig. 25).

We conducted transient photocurrent and transient photovoltage (Supplementary Fig. 26) measurements, which indicate that the

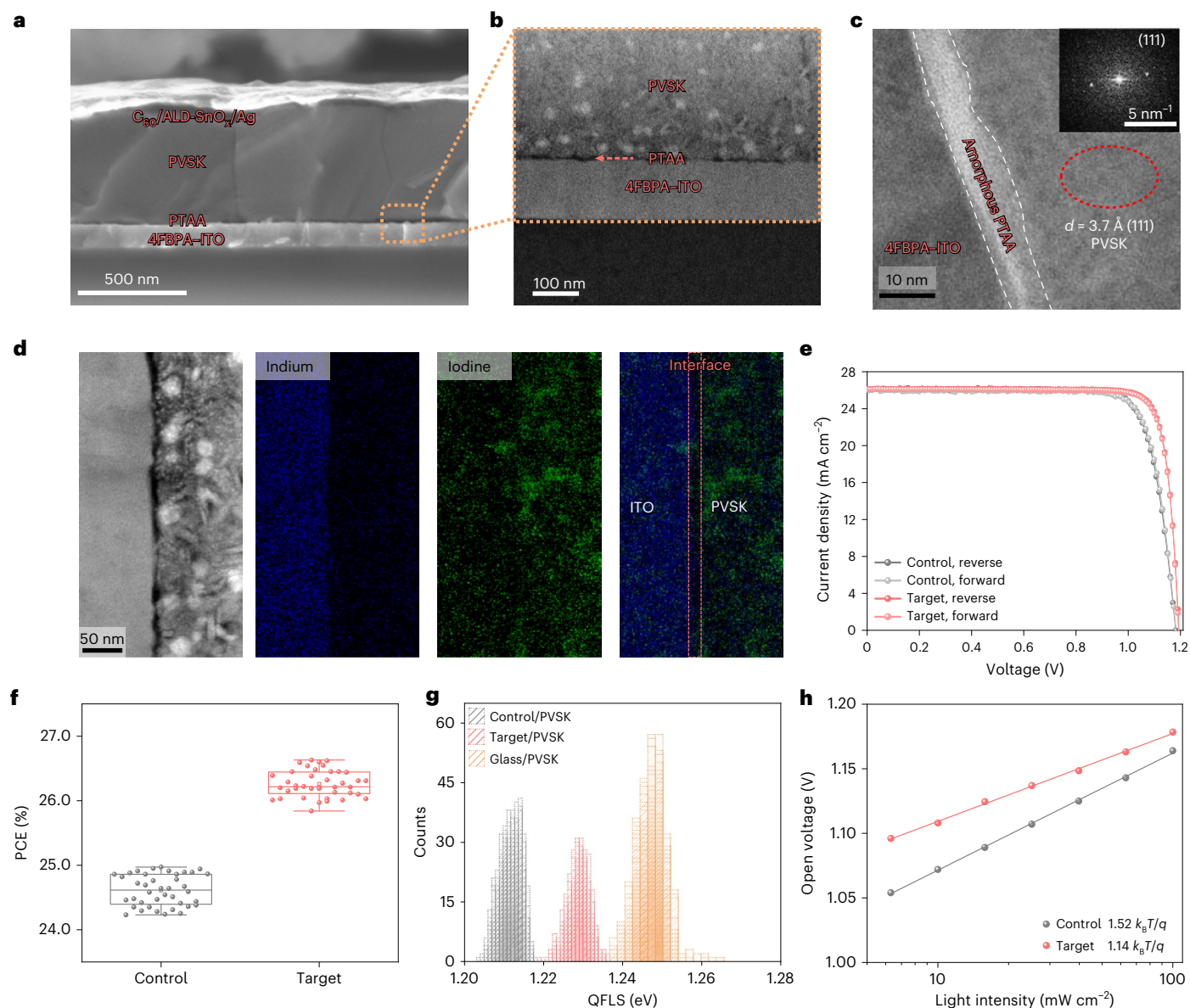


Fig. 3 | Impact of 4FBPA-ITO/amorphous PTAA on perovskite films.

a, Cross-sectional SEM image of the target PSC device. **b**, Cryo-STEM image of the target stack. **c**, Zoom-in cryogenic transmission electron microscopy (cryo-TEM) image of the target stack with the amorphous PTAA, along with the (111) crystal plane of the perovskite obtained using fast Fourier transform. **d**, Zoom-in cryo-STEM images of the 4FBPA-ITO/PTAA/perovskite stack with the characteristic element maps for indium (ITO) and iodine (perovskite). Note: The bright dots in the cryo-STEM image result from the high dose of electron beam for perovskite degradation to PbI_2 . However, this will not affect the evaluation of

the 4FBPA-ITO/PTAA/perovskite stack. **e**, Forward and reverse current-voltage (J - V) scans of the best control and target p-i-n PSCs under 1-sun illumination. **f**, Statistical PCE distribution for 80 devices (40 control devices and 40 target devices). The central line of the box plot represents the median, the box length indicates data dispersion and the whiskers denote the maximum and minimum values. **g**, QFLS of glass/perovskite, control PTAA/perovskite and target PTAA/perovskite stacks. **h**, Light intensity versus open voltage plots under 1 mW cm^{-2} to 100 mW cm^{-2} (1% to 100% light intensity) LED illumination.

target device has superior charge carrier extraction compared to the control device. Thus, the enhanced fill factor (FF) can be attributed to the efficient hole extraction at the ITO/PTAA interface, as well as lower series resistance of amorphous PTAA on 4FBPA-ITO substrate³² (Supplementary Fig. 27 and Supplementary Table 5). Moreover, we performed quasi-Fermi level splitting (QFLS) measurements (Fig. 3g and Supplementary Fig. 28) on the control and target perovskite stacks. The larger splitting in the target stack indicates reduced open-circuit voltage (V_{oc}) loss at the bottom interface³³. The higher QFLS values of the target perovskite films are directly correlated with the higher V_{oc} of the target device. We calculated the ideality factors (n_{id}) for the control and target devices to examine the relationship between V_{oc} and light intensity (Fig. 3h). The lower n_{id} value ($n_{id} = 1.14$) for the target device

suggests an effective suppression of non-radiative recombination compared to the control device³⁴. This series of electrical analyses provides solid evidence for the enhanced V_{oc} of the target devices, which can be primarily attributed to effective suppression of non-radiative recombination³⁵ and an increase in the QFLS within the target device.

The incident photon-to-electron conversion efficiency and the corresponding integrated short-circuit photocurrent density (J_{sc}) for both the control and target devices are presented in Supplementary Fig. 29. The integrated J_{sc} matches the values of the previous J - V curve. Specifically, the target device exhibits higher J_{sc} due to the higher level of photogenerated carriers (Supplementary Fig. 20) and the efficient extraction at the 4FBPA-ITO and PTAA interface. The fabrication process of p-i-n PSMs follows the flow chart in Fig. 1a. The reverse scan

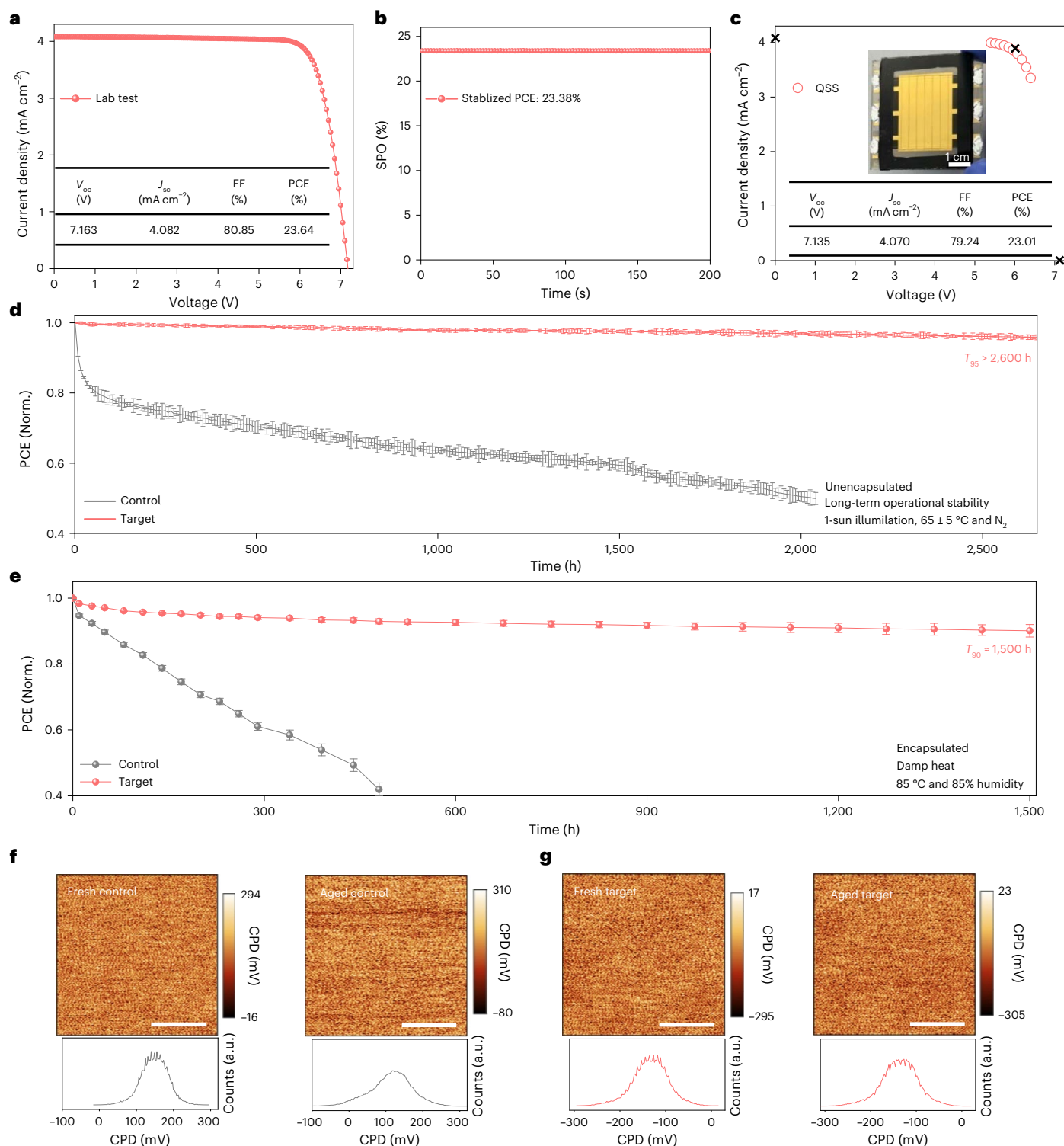


Fig. 4 | Photovoltaic performance and stability measurements. **a**, J - V reverse curve of the best-performing target PSM. **b**, Stabilized power output of the target PSM. **c**, QSS J - V curve and maximum power output point of a representative 4FBPA-ITO/PTAA-based 6.15 cm² PSM device (insert), certified by the National Renewable Energy Laboratory (NREL) using asymptotic maximum power scan protocols. Note: for third-party certification, the device used Au electrodes instead of the Ag electrodes used in laboratory experiments. **d**, Normalized (Norm.) PCE with error bars for the unencapsulated control and target PSMs (four

devices each) measured at maximum-power-point tracking under continuous 1-sun solar illumination, 65 ± 5 °C and nitrogen atmosphere. **e**, Normalized PCE with error bars for the encapsulated control and target PSMs (four devices each) measured under damp-heat conditions (85 °C and 85% humidity). Note: error bars in **d** and **e** represent the standard deviation calculated from the measurement of four samples in each case. **f, g**, Surface potential images acquired by KPFM of the control (**f**) and target (**g**) PTAA before and after aging at 85 °C for 1,000 hours. Scale bar, 0.5 μm.

efficiency reaches 23.64% (Fig. 4a), and the stabilized power output for the PSM is 23.38% (Fig. 4b). We also obtained a certified QSS efficiency of 23.01% (Fig. 4c and Supplementary Fig. 30).

Additionally, we conducted stability measurements to evaluate the operational and damp-heat stability of both the control and target PSMs. As shown in Fig. 4d, the unencapsulated target PSM maintained 95.9% of its original efficiency over 2,600 hours of continuous maximum-power-point operation under 1-sun illumination in a N₂ atmosphere at 65 ± 5 °C. This represents the best operational stability achieved to date among all reported perovskite modules^{36–45} (Supplementary Table 6). By contrast, the control device exhibited a loss of approximately 54% in efficiency after 2,000 hours of aging under the same conditions. The results of damp-heat stability testing (Fig. 4e) exhibit a similar trend, with the encapsulated target device showing only 10% efficiency loss after a 1,500-hour damp-heat test at 85 °C and 85% relative humidity, and the control devices experiencing decreases of ~60% in initial efficiency after 480 hours. The variations in the corresponding V_{oc} , J_{sc} , and FF parameters for both the control and target devices in the maximum-power-point tracking and damp-heat measurements are shown in Supplementary Fig. 31.

To explore the factors behind the excellent stability of the target devices, we measured the CPD values of ITO/PTAA and 4FBPA-ITO/PTAA films (Fig. 4f,g) before and after aging for 1,000 hours at 85 °C. The CPD of the 4FBPA-ITO/PTAA film is almost unchanged after aging, while the CPD distribution range of the ITO/PTAA film becomes wider. We peeled off the perovskite films from both fresh and aged control (bare ITO/PTAA) and target (4FBPA-ITO/PTAA) substrate stacks. Then, we compared their morphology through SEM and PL hyperspectral mapping images. According to the peeled-off SEM images (Supplementary Fig. 32), the perovskite peeled from the 4FBPA-ITO/PTAA substrate contains more PTAA residue, indicating a stronger adsorption force and better contact for the perovskite within the 4FBPA-ITO/PTAA stack. In the aged samples the control film shows less residual PTAA, suggesting degradation at the ITO/perovskite interface during the aging process, while the residual PTAA on the target film changes slightly, indicating a distinct PTAA morphology that impacts the interfacial contacts between the perovskite and ITO photoanode. Additionally, the aged target perovskite exhibits uniform and slightly reduced PL intensity compared to the fresh target samples. By contrast, the aged control perovskite shows scattered yellow dots with lower PL intensity than the fresh control sample (Supplementary Fig. 33). Taken together, our findings suggest that the well-aligned energy levels between 4FBPA-ITO and PTAA, combined with the uniform amorphous PTAA and the robust interface between PTAA and perovskite, are beneficial for advancing the efficiency and operational stability of state-of-the-art perovskite photovoltaics.

Conclusions

In this work, we demonstrated that surface modification of ITO with amphiphilic 4FBPA introduces nanoscale amorphization of PTAA, leading to an NREL-certified QSS efficiency of 23.01% in perovskite modules. We identify the governing factors of PTAA film formation and establish a clear correlation between PTAA morphology and device performance. Notably, PTAA aggregation and its intrinsically weak crystallization are revealed as key bottlenecks limiting device efficiency. This concept can be extended to functional organic materials for precise morphology control of polymeric HSLs, thereby advancing the performance of perovskite photovoltaics.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41565-026-02207-z>.

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Methods

Materials

The following chemicals were used in the experiments: formamidinium iodide (FAI, 99.99%, Greatcell Solar Materials); methylammonium chloride (MACI, 99.99%, Greatcell Solar Materials); caesium iodide (CsI) (99.999%, ultra dry, TCI Co. Ltd); lead iodide (PbI_2 , 10-mesh beads, 99.999%, ultra dry, Alfa Aesar); 4-*tert*-butylpyridine (tBP, 98%, Sigma-Aldrich); lithium bistrifluorosulfonyl imide (Li-TFSI, Sigma-Aldrich); potassium ferricyanide (99%, Sigma-Aldrich); anhydrous dimethylformamide (DMF, 99.8%, Sigma-Aldrich); *N*-methyl-2-pyrrolidone (NMP, 99.7%, Sigma-Aldrich); toluene (99.9%, Sigma-Aldrich); chlorobenzene (CB, 99.8%, Sigma-Aldrich); isopropanol (IPA, 99.8% ACROS Organics); acetone (99.8%, ACROS Organics); 2-methoxyethanol (2-Me, Sigma-Aldrich); oleylamine (Sigma-Aldrich) L- α -phosphatidylcholine (LP, Sigma-Aldrich); fullerene (C_{60} , 99.95%, Nano-C); benzylphosphonic acid (BPA, Sigma-Aldrich); 4-fluorobenzylphosphonic acid (4FBPA) (Sigma-Aldrich); poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA, MW 3,000–6,000 g mol^{-1}), propylammonium chloride (PACl) and 4-fluoro-phenethylammonium bromide (4F-PEABr) (Xi'an Polymer Light Technology); and ITO ($15 \Omega \text{ sq}^{-1}$) substrates (Xin Yan Technology Limited).

Unless otherwise specified, all materials were used as received.

Perovskite film and fabrication of PSC and PSM devices

The patterned ITO substrates were sequentially washed using Hellmanex (2% in deionized water), acetone and isopropanol, with each step involving 15 min of sonication. After drying with compressed air, the ITO substrates were subjected to a 15-min UV-ozone treatment. In the ITO photoanode modification process, the organic phosphonic acids (BPA and 4FBPA) were dissolved in IPA and CB (volume ratio = 1:9) at a concentration of 0.4 mg ml^{-1} ; then precursor solution was blade-coated onto the cleaned ITO glass substrates with a gap of $50 \mu\text{m}$, with a speed of 15 mm s^{-1} . During blade coating, the N_2 knife was operated at 10 bars of pressure. After that, the coated ITO substrates were annealed at 100°C for 10 min to form BPA-ITO and 4FBPA-ITO. For the hole transport layer deposition, PTAA dissolved in CB at a concentration of 1.5 mg ml^{-1} was blade-coated onto the ITO or OPA-ITO surfaces at a speed of 15 mm s^{-1} . During blade coating, the N_2 knife was operated at 10 bars of pressure. Then, the coated substrates were annealed at 100°C for 10 min.

Perovskite precursor ink was prepared by dissolving a mixture of PbI_2 (796.6 mg), FAI (256.6 mg), MACI (21.2 mg), PACl (15.04 mg), CsI (20.4 mg), LP (0.27 mg) and oleylamine (2 μl) in 1 ml of a solution containing 2-Me (500 μl) and DMF (500 μl), followed by the addition of NMP (72 μl) and shaking until complete dissolution. The perovskite precursor was applied to the prepared PTAA substrate using a N_2 flow-assisted blade-coating process. The perovskite precursor solution was blade-coated onto the PTAA substrates with a gap of $200 \mu\text{m}$ at a speed of 10 mm s^{-1} . During blade coating, the N_2 knife was operated at 15 bars of pressure. The substrate was then annealed at 120°C for 30 min. Amounts of 1 mg 4F-PEABr and 3 mg MACI were dissolved in 1 ml IPA (with 1% DMF) and the resulting solution was blade-coated onto the perovskite with a gap of $50 \mu\text{m}$ at a speed of 15 mm s^{-1} . During blade coating, the N_2 knife was operated at 10 bars of pressure, followed by annealing at 100°C for 5 min. Then, thermal evaporation and atomic layer deposition (ALD) were used to sequentially deposit layers of C_{60} (20 nm, $0.2 \text{ \AA s}^{-1}$) and SnO_x (20 nm), respectively. Finally, 100 nm Ag was deposited at a rate of $0.8 \text{ \AA s}^{-1}$ using thermal evaporation.

Standard thin-film scribing steps for PSMs

Standard thin-film photovoltaic scribing processes (P1, P2, P3 and P4) were employed for mini-module fabrication. All laser scribing steps were performed using a 532-nm nanosecond laser. The cleaned glass/ITO substrates ($5 \times 5 \text{ cm}^2$) were first laser scribed to create P1 with width

of $25 \mu\text{m}$. After the deposition of the OPA, PTAA, perovskite layer, C_{60} and ALD- SnO_x layer, the samples were scribed to form P2 with a width of $35 \mu\text{m}$. After deposition of 100-nm Ag electrode, the samples were scribed to form P3 with a width of 40 nm. Finally, the modules were formed by etching procedure (P4, 40 nm) to electrically isolate the edges of the mini-modules.

TEM and EELS characterization

Cross-sectional TEM samples were prepared using a focused ion beam system (Helios G4 UX) equipped with a cryo-autoloader. Cryogenic high-resolution TEM images were acquired at 80 K using an aberration-corrected FEI (Themis Z) at 300 kV. Cryo-EELS spectra were acquired using a 15-pA current and 0.05-s dwell time at each pixel.

Photovoltaic performance measurements

Unless specifically mentioned, all devices were characterized in an unencapsulated state and within a N_2 atmosphere, using a solar simulator integrated into a glove box. The J - V curves of the PSCs were measured using a Keithley 2400 Source Meter under a simulated 100 mW cm^{-2} spectrum (AM 1.5 G) from a Xenon-lamp-based solar simulator (Abet Technologies Sun 3000 Class AAA Solar Simulator). For the small-area device, the bias voltage was applied from 1.2 V to -0.1 V , while for the module device it was swept from 7.5 V to -0.1 V . The mask area was individually measured under an optical microscope. The light intensity was calibrated using a commercial silicon reference cell certified by the NREL, with the aid of a Scott visible-colour glass filter. The fabricated devices were directly measured without any encapsulation. External quantum efficiency was measured using a focused, 200-Hz chopped monochromatic beam from a Xenon lamp.

Optical measurements

Femtosecond transient absorption measurements were performed using a Helios spectrometer (Ultrafast System). For this purpose, the samples were excited with 450-nm pump pulses generated by directing a fraction of the 800-nm output from an Astrella laser (Coherent; ~ 150 -fs pulse, 1 kHz, 7 mJ pulse $^{-1}$) into the spectrally tunable optical parametric amplifier (Newport Spectra-Physics). Excitation power was adjusted to 13 μW using neutral density filters. The probe pulses (UV visible and NIR wavelength continuum, white light) were generated by passing another fraction of the 800-nm pulses through a 2-mm-thick sapphire crystal. Before white-light generation, the 800-nm pulses were passed through a motorized delay stage. Depending on the movement of the delay stage, transient species were detected following excitation at different time scales. The white light was split into two beams (signal and reference) and focused on two fibre optics to improve the signal-to-noise ratio. The excitation pump pulses were spatially overlapped with the probe pulses on the samples after passing through a synchronized mechanical chopper (500 Hz) that blocked alternative pump pulses. Changes in absorption (ΔA) were recorded as a function of time delay and wavelength, enabling the study of transient species over various time scales and spectral regions. All data were fitted using Surface Explorer software (Ultrafast System).

Steady-state PL was measured using an Edinburgh FLS920 Fluorescence Spectrometer equipped with a 450-W continuous-wavelength Xenon lamp. Hyperspectral luminescence images were obtained using a 50-fold-objective optical microscope under 450-nm excitation from a pulse laser, with the excitation intensity maintained below 1×10^6 counts. Photoluminescence quantum yield was measured using an Enlitech LQ-100X system within an integrated sphere under 405-nm excitation.

Time-dependent PL spectra measurement and QFLS calculation

For time-dependent PL spectra measurement, perovskite films on bare glass were encapsulated in a nitrogen glove box using an ultra-thin glass

cover slide with an edge sealed by UV-curable epoxy. The PL spectra were recorded approximately every minute using a hyperspectral luminescence imaging system equipped with an optical microscope with a $\times 20$ objective and a 532-nm laser with controlled laser intensity.

QFLS ($\Delta\mu$) was calculated based on the generalized Planck emission law, described as follows:

$$\Phi_{\text{PL}}(E) = \frac{2\pi A(E) E^2}{h^3 c^2} \frac{1}{\exp\left[\frac{E - \Delta\mu}{k_B T}\right] - 1} \approx \frac{2\pi A(E) E^2}{h^3 c^2} \exp\left[-\frac{E}{k_B T}\right] \exp\left[\frac{\Delta\mu}{k_B T}\right] \quad (1)$$

where h is the Planck constant, c is the speed of light in vacuum, k_B is the Boltzmann constant and T is the absolute temperature. In the high-energy emission region ($E > E_g$), the absorptivity $A(E)$ could be acquired from the following equation:

$$A(E) = \frac{(1 - R)(1 - \exp(-\alpha d))}{1 - R \exp(-\alpha d)} \quad (2)$$

Where α is the absorption coefficient, d is film thickness and R is reflectivity.

The above equation can be converted to:

$$\ln\left(\frac{\Phi_{\text{PL}}(E) h^3 c^2}{A(E) 2\pi E^2}\right) = -\frac{E}{k_B T} + \frac{\Delta\mu}{k_B T} \quad (3)$$

The QFLS and T (absolute temperature) can be determined by extracting the intercept and slope from the curves of equation (3) plotted against E . By fitting the PL spectra of each pixel at the high-energy emission region, we can transform the absolute PL image into a QFLS distribution map. A homemade MATLAB program was employed to execute the conversion process. To avoid any graphic noise, 5×5 point averaging was applied.

Electronic and chemical properties of the ITO photoanode

X-ray photoelectron spectroscopy studies were carried out on a Kratos Axis Supra spectrometer equipped with a monochromatic Al K α X-ray source ($h\nu = 1486.6$ eV) operating at 75 W and a multi-channel plate and delay-line detector, under a vacuum of $\sim 10^{-9}$ mbar. All spectra were recorded using an aperture slot of 300×700 μm . Survey spectra were collected using a pass energy of 160 eV and a step size of 1 eV. A pass energy of 20 eV and a step size of 0.1 eV were used for the high-resolution spectra. UV photoelectron spectroscopy measurements were performed on the same instrument using the helium I ($h\nu = 21.22$ eV) photon source.

Crystallographic and morphological characterization

GIWAXS was measured using a Xenocs Xeuss 3.0 (XENOCs) with a Cu source (wavelength $\lambda = 1.54178$ Å) at room temperature. The incident angle was 0.2° ; the sample-to-detector distance was 72 mm. X-ray diffraction patterns were collected using a Bruker AXSD8 diffractometer with Cu K α radiation (slit width = 0.6 mm, $\lambda = 1.54178$ Å).

SEM images were obtained using a Helios G4 UX dual-beam scanning electron microscope. KPFM and current-sensitive atomic force microscopy measurements, including relevant potential, surface current and topographic images, were obtained using a Digital Instruments Multimode AFM (Veeco Metrology Group) with specific AFM tips (model SCM-PIT; nominal spring constant $1\text{--}5$ N m $^{-1}$; front and back sides coated with 20 nm PtIr). All samples were grounded during the potential measurements. Additionally, a scan rate of 0.5 Hz and a -0.5 V DC bias were applied during the current-sensitive atomic force microscopy measurement.

Electrical measurements

Light-intensity dependence, transient photovoltage and transient photocurrent measurements were performed using a commercial

Fluxim AG platform. Unencapsulated devices were exposed to modulated light from a standard white LED with the intensity ranging from 1% to 100% over a frequency range of 0.01 Hz to 100 kHz. Electrochemical impedance spectroscopy measurements were carried out under open-circuit voltage conditions for the devices in the dark, across a frequency range from 10 Hz to 100 kHz. Transient photovoltage measurements were tested under open-circuit conditions, while transient photocurrent measurements were obtained using flashing devices with a continuous light pulse (0.1% to 100% intensity, 100- μs pulse length) under short-circuit conditions.

Computational methods

DFT calculations were carried out using the projector-augmented wave method as implemented in the Vienna Ab initio Simulation Package. The generalized gradient approximation together with the Perdew–Burke–Ernzerhof exchange–correlation functional was used. The van der Waals interactions were also included in the DFT calculations using Grimme’s zero damping DFT-D3 method. The crystal structure of bulk ITO was constructed following the methods outlined in a previous study⁴⁶. The ITO slab was built with a 2×2 lateral periodicity and an exposed (100) surface; a ~ 20 -Å vacuum was introduced to prevent interactions between slab replicas. A uniform $4 \times 4 \times 4$ k -mesh was used in the Brillouin zone to optimize the crystal structure of bulk ITO; a $4 \times 4 \times 1$ k -mesh was used for ITO (100) slab, and a $2 \times 2 \times 1$ k -mesh was used for the PTAA/ITO (100), PTAA/BPA–ITO (100) and PTAA/4FBPA–ITO (100) interfaces. The energy cut-offs of wave functions were set at 500 eV for the bulk, 450 eV for the slab and 400 eV for the interfaces. The atomic positions of all structures were fully relaxed until the Hellmann–Feynman forces on each atom were less than 0.015 eV Å $^{-1}$.

Cyclic voltammetry methods

In the electrochemical measurements, the surface-passivation (blocking) effect of the OPAs on ITO was evaluated using a redox-probe approach. Cyclic voltammetry was performed in an aqueous electrolyte containing 0.1 M KCl as the supporting electrolyte and 10 mM potassium ferricyanide ($\text{K}_3[\text{Fe}(\text{CN})_6]$) as the redox probe at a scan rate of 100 mV s $^{-1}$. Four consecutive cycles were recorded for each sample; only the last three cycles were used for data analysis. A conventional three-electrode configuration was employed with ITO (bare or OPA-modified) as the working electrode, a Pt wire as the counter electrode and an Ag/AgCl electrode as the reference. All solutions were prepared with deionized water, and measurements were carried out at room temperature. The shielding (blocking) efficiency of the OPAs was assessed by comparing the redox-probe cyclic voltammetry responses obtained on OPA-modified ITO with those on bare ITO under identical conditions.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available in the main text or Supplementary Information. Correspondence and requests for materials should be addressed to the corresponding authors (S.Y., K.Z. and O.M.B.).

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- Li, H., Paramonov, P. & Bredas, J.-L. Theoretical study of the surface modification of indium tin oxide with trifluorophenyl phosphonic acid molecules: impact of coverage density and binding geometry. *J. Mater. Chem.* **20**, 2630–2637 (2010).

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Author contributions

H.Z., B.S. and Z.S. contributed equally to this work. O.M.B. and K.Z. directed and supervised the research. O.M.B., K.Z., H.Z., B.S., S.Y. and Z.S. conceptualized and designed the experiments. H.Z., B.S. and S.Y. performed device fabrication, characterization and evaluation. J.Y. carried out the DFT calculations. S.Z. calculated the QFLS. R.Z. conducted the SEM and energy-dispersive X-ray spectroscopy (EDS) characterizations. M.N.H. performed the X-ray photoelectron spectroscopy and UV photoelectron spectroscopy measurements. Y.Y. characterized the GIWAXS. Q.W. conducted the cryo-focused ion beam (FIB)-TEM and EELS measurements and analysis. L.G.-A. conducted time-resolved PL and femtosecond transient absorption spectroscopy measurements. M.A., A.J., I.G., M.F. and O.F.M. provided guidance and expertise. H.Z., B.S., S.Y., Z.S., K.Z. and O.M.B. wrote the paper. All authors participated in discussions of the results and paper review.

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Competing interests

O.M.B. is a founder of Quantum Solutions, a company specializing in the development of optoelectronic devices. The other authors declare no competing interests.

Additional information

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells	☒ Yes	Small-area devices: 0.0996 cm ² ; Module devices: 6.5 cm ²
	☐ No	Explain why this information is not reported/not relevant.
Method used to determine the device area	☒ Yes	The information can be found in Photovoltaic performance measurements part
	☐ No	Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes	As shown in Fig. 3e and Table 1
	☐ No	
Voltage scan conditions	☒ Yes	Photovoltaic performance measurements section
	☐ No	Explain why this information is not reported/not relevant.
Test environment	☒ Yes	Photovoltaic performance measurements section
	☐ No	Explain why this information is not reported/not relevant.
Protocol for preconditioning of the device before its characterization	☐ Yes	Provide a description of the protocol.
	☒ No	We did not use any preconditioning
Stability of the J-V characteristic	☒ Yes	As shown in Fig. 4b and Fig. S24
	☐ No	Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes	Provide a description of hysteresis or any other unusual behaviour observed during the characterization.
	☒ No	No unusual behavior was observed, with negligible hysteresis
Related experimental data	☒ Yes	Fig. 3e, Fig. S22b, and Table S4
	☐ No	Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes	The EQE was measured using a focused, 200 Hz-chopped monochromatic beam from a Xenon lamp.
	☐ No	Explain why this information is not reported/not relevant.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes	Fig. S29
	☐ No	Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	Provide a description of the measurement conditions. No tandem cells investigated
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Photovoltaic performance measurements section Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cell was calibrated by NREL Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	Photovoltaic performance measurements section Explain why this information is not reported/not relevant.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Small-area devices: 0.07973 cm² (Fig. S23); Module devices: 6.15 cm² (Fig. S30) Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. We measured all devices with masks
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	National Renewable Energy Laboratory (NREL) Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	Fig. S23 and Fig. S30 Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	40 control devices and 40 target devices (Fig. 3f) Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Fig. 3f Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Fig. 4e and Fig. 4d Explain why this information is not reported/not relevant.